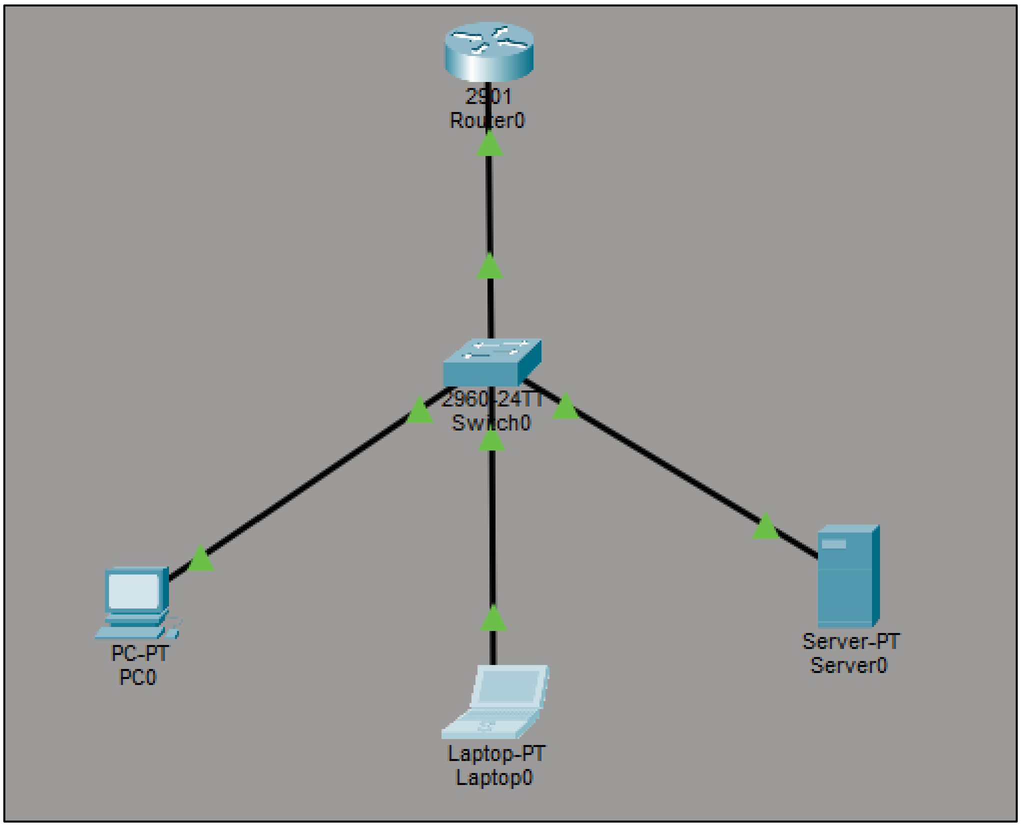
1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).



2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Component	Quantity	Brand/Model Suggestion	Purpose
Wi-Fi Router	1	TP-Link Archer AX55 / ASUS RT-AX58U	Connects the local network to the Internet and provides Wi-Fi
Modem/ONT	1	ISP-provided modem/ONT	Connects the network to the ISP
Gigabit Switch	1	TP-Link TL-SG108 (8-port)	Connects multiple wired computers/devices
Wireless Access Point	1 (optional)	TP-Link EAP225 / EAP610	Extends Wi-Fi coverage
Ethernet Cable	As required	Cat6 UTP cable	Connects router, switch, PCs, printer, AP, etc.
RJ45 Connectors	As required	Cat6 RJ45 connectors	Used to terminate Ethernet cables
Network Adapter	1 per wired device if needed	TP-Link USB-to-Ethernet adapter	Provides Ethernet connectivity to devices without an RJ45 port
Wi-Fi Extender	0–1 (optional)	TP-Link RE505X	Extends Wi-Fi to areas with weak signal
Network Printer	1 (optional)	HP LaserJet Pro series	Allows multiple users to print over the network
UPS	1	APC Back-UPS 600/650VA	Provides backup power to router, modem and switch
Patch Panel	1 (optional)	8/12-port Cat6 patch panel	Organizes permanent Ethernet connections
Network Rack	1 (optional)	6U wall-mount rack	Holds and organizes networking equipment

3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Step-by-step: Configure a Wi-Fi Router for a SOHO Network

Connect the hardware

- Connect the ISP modem/ONT to the router's **WAN/Internet port** using an Ethernet cable.
- Connect your laptop/PC to a **LAN port** on the router or connect using the router's default Wi-Fi.

Open the router's settings

- Open a web browser.
- Enter the router's management address, such as 192.168.0.1 or 192.168.1.1.
- Log in using the default administrator username and password shown on the router label/manual.

Change the default admin password

- Go to **Administration / System / Management**.
- Find **Administrator Password**.
- Replace the default password with a strong unique password.
- Save/apply the changes.

Create a unique SSID

- Go to **Wireless / Wi-Fi Settings**.
- Change the default network name (SSID).
- Example: Dhrumit_SOHO
- Avoid using personal information such as your full name, phone number, or address.

Enable WPA2 security

- Open **Wireless Security** settings.
- Select **WPA2-Personal (WPA2-PSK)**.
- Choose **AES** encryption if the router provides a separate encryption option.
- Create a strong Wi-Fi password, for example: SOHO@Net_2026!Secure

Configure both Wi-Fi bands

- If the router supports **2.4 GHz and 5 GHz**, configure both.
- You can use separate SSIDs such as:

- Dhrumit_SOHO_2.4G
- Dhrumit_SOHO_5G

Save the configuration

- Click **Save/Apply**.
- The router may restart and disconnect your device temporarily.

Reconnect your devices

- Search for the new SSID on your laptop, phone, printer, etc.
- Enter the new Wi-Fi password.
- Confirm that the devices receive an IP address and can access the Internet.

Verify security

- Return to the router's Wi-Fi settings.
- Confirm that **WPA2-Personal** is enabled.
- Confirm that the default administrator password is no longer being used.

4. Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

```
Device Name: Switch0
Custom Device Model: 2960 IOS15
Hostname: Switch

Port                Link   VLAN   IP Address          MAC Address
FastEthernet0/1     Up     1      --                  00E0.F7E9.A901
FastEthernet0/2     Up     1      --                  00E0.F7E9.A902
FastEthernet0/3     Down   1      --                  00E0.F7E9.A903
FastEthernet0/4     Up     1      --                  00E0.F7E9.A904
FastEthernet0/5     Down   1      --                  00E0.F7E9.A905
FastEthernet0/6     Down   1      --                  00E0.F7E9.A906
FastEthernet0/7     Down   1      --                  00E0.F7E9.A907
FastEthernet0/8     Down   1      --                  00E0.F7E9.A908
FastEthernet0/9     Down   1      --                  00E0.F7E9.A909
FastEthernet0/10    Down   1      --                  00E0.F7E9.A90A
FastEthernet0/11    Down   1      --                  00E0.F7E9.A90B
FastEthernet0/12    Down   1      --                  00E0.F7E9.A90C
FastEthernet0/13    Down   1      --                  00E0.F7E9.A90D
FastEthernet0/14    Down   1      --                  00E0.F7E9.A90E
FastEthernet0/15    Down   1      --                  00E0.F7E9.A90F
FastEthernet0/16    Down   1      --                  00E0.F7E9.A910
FastEthernet0/17    Down   1      --                  00E0.F7E9.A911
FastEthernet0/18    Down   1      --                  00E0.F7E9.A912
FastEthernet0/19    Down   1      --                  00E0.F7E9.A913
FastEthernet0/20    Down   1      --                  00E0.F7E9.A914
FastEthernet0/21    Down   1      --                  00E0.F7E9.A915
FastEthernet0/22    Down   1      --                  00E0.F7E9.A916
FastEthernet0/23    Down   1      --                  00E0.F7E9.A917
FastEthernet0/24    Down   1      --                  00E0.F7E9.A918
GigabitEthernet0/1  Up     1      --                  00E0.F7E9.A919
GigabitEthernet0/2  Down   1      --                  00E0.F7E9.A91A
Vlan1               Down   1      <not set>           0010.11E3.144A
```

5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Basic troubleshooting commands

Open **Command Prompt** and run:

- ipconfig

Check whether the laptop has a valid IP address.

Then:

- ping 192.168.1.1
- **Reply received:** The laptop can communicate with the router.
- **Request timed out:** Check Wi-Fi connection, IP configuration, or router connection.

Next:

- ping 8.8.8.8

Reply received: Internet connectivity is working; DNS may be the problem.

No reply: Check the router, ISP connection, or laptop network configuration.

Finally:

- ping google.com

If 8.8.8.8 works but google.com does not, investigate **DNS settings**.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=20ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 20ms, Average = 5ms
```